

Community Solar & Credit Sharing Platform

System Requirements Specification Matrix

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional ▾	The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.	High ▾	Pass: Solar generation is recorded and a valid credit transfer deducts credits from the prosumer and applies them to the neighbor's account. Fail: A transfer exceeding the available credit balance is rejected.	Core functionality for recording solar generation and sharing surplus credits.
FR-002	Functional ▾	The system shall allow a Prosumer Resident to view their current solar credit balance and monthly solar generation details.	High ▾	Pass: The system displays the current credit balance and generation data for the selected month. Fail: The system displays an incorrect or unavailable balance.	Allows residents to monitor their generated energy and available credits.
FR-003	Functional ▾	The system shall validate a solar credit transfer by verifying the recipient account and the sender's available credit balance before completing the transfer.	High ▾	Pass: A transfer is completed only when the recipient exists and sufficient credits are available. Fail: An invalid recipient or insufficient balance prevents the transfer.	Prevents invalid or unauthorized credit transfers.
FR-004	Functional ▾	The system shall apply available solar credits to a Prosumer Resident's monthly utility billing statement and	High ▾	Pass: Applied credits are deducted from the available balance and the corresponding billing offset is	Connects community solar credits with monthly utility billing.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
		calculate the resulting billing offset.		shown. Fail: The system does not apply credits when the available balance is insufficient.	
FR-005	Functional ▾	The system shall allow the Co-op Manager to generate monthly reports containing household solar generation, credit transfers, and billing offsets.	Medium ▾	Pass: A monthly report containing solar generation, credit transfers, and billing offsets is generated. Fail: The report cannot be generated when required monthly data is unavailable.	Helps the Co-op Manager monitor community solar activity.
NFR-001	Nonfunctional — Perfor... ▾	The smart meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.	High ▾	Pass: Testing confirms that hourly generation data for 500 households is processed while maintaining at least 99.9% uptime. Fail: The system falls below the 99.9% uptime requirement or fails to process the required data.	Ensures reliable operation when supporting many households.
NFR-002	Nonfunctional — Security ▾	The system shall protect solar credit and billing information so that only authenticated and authorized users can access or modify it.	High ▾	Pass: Unauthorized users are prevented from accessing or modifying credit and billing information during security testing. Fail: An unauthorized user successfully accesses or modifies the information.	Protects sensitive credit and billing information from unauthorized access.